

ORIGINAL ARTICLE

The association between the incidence of intestinal graft-vs-host disease and antibiotic use after allogeneic hematopoietic stem cell transplantation

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Abstract

Intestinal microbiota plays an important role in the regulation of allogeneic immune reaction after allogeneic hematopoietic stem cell transplantation (allo-SCT). Intestinal graft-vs-host disease (GVHD) is one of the major causes of mortality after allo-SCT and often complicated with intestinal dysbiosis. Recent studies suggest that antibiotic-induced dysbiosis is a risk factor for intestinal GVHD. We retrospectively evaluated the impacts of antibiotic use on the incidence of intestinal GVHD occurring before day 100 after allo-SCT. Among 213 patients who underwent allo-SCT, 200 patients achieving engraftment were analyzed. Antibiotics were classified into carbapenem, quinolone, penicillin, cephem, and glycopeptide. Among 128 patients who developed acute GVHD, intestinal GVHD developed in 36 patients. Patients with intestinal GVHD received significantly longer administration of carbapenem and glycopeptide compared to those without it in periengraftment period. In multivariate analysis, use of carbapenem for greater than 7 days was associated with an increased risk of intestinal GVHD. However, use of antibiotics for greater than 7 days was not associated with poor overall survival and high nonrelapse mortality. Long use of carbapenem in periengraftment period may be a risk for intestinal GVHD. Prospective studies are required to validate our findings.

KEYWORDS

antibiotic, graft- vs-host disease (GVHD), microbiomics

1 | INTRODUCTION

Intestinal graft-vs-host disease (GVHD) is one of the major causes of mortality after allogeneic hematopoietic stem cell transplantation (allo-SCT). We and others have shown that intestinal stem cells and Paneth cells are targeted by GVHD, and injury of Paneth cells results in impaired secretion of the antimicrobial peptides such as α -defensins into the lumen and dysbiosis.¹⁻⁴ Intestinal microbiota plays an important role in the regulation of allogeneic immune reaction

after allo-SCT.^{2,4,5} Emerging data indicated that dysbiosis is associated with intestinal GVHD.^{2,6} In addition, a recent study showed that the abundance of specific intestinal bacteria was associated with relapse after allo-SCT independent on intestinal GVHD.⁷

Prophylaxis and treatment of bacterial infections are important for success of allo-SCT.⁸ Prolonged neutropenia and conditioning-mediated mucosal injury are risks for bacterial infections in early phase of allo-SCT, and systemic administration of broad-spectrum antibiotics is often used. However, broad-spectrum antibiotics also

TABLE 1 Patient characteristics

Total no. of patients	n = 200
Age, median (range)	45 (17-67)
Male sex	116 (58.0%)
Diagnosis	
AML	75 (37.5%)
ALL/LBL	41 (20.5%)
ML/ATLL	42 (21.0%)
MDS	26 (23.0%)
CML	5 (2.5%)
AA	4 (2.0%)
Graft source	
BM match	57 (28.5%)
BM mismatch	34 (17.0%)
PBSC match	25 (12.5%)
PBSC mismatch	39 (19.5%)
CB	45 (22.5%)
Conditioning regimen	
Myeloablative	90 (45.0%)
Reduced intensity	110 (55.0%)
GVHD prophylaxis	
CSP/TAC + sMTX	136 (68.0%)
CSP/TAC + MMF	16 (8.0%)
CSP/TAC + MMF + PTCy	21 (10.5%)
CSP/TAC alone	18 (9.0%)
CSP/TAC + sMTX + ATG	9 (4.5%)

AML, acute myeloid leukemia; ALL, acute lymphoblastic leukemia; LBL, lymphoblastic leukemia; ML, malignant lymphoma; ATLL, adult T-cell leukemia/lymphoma; MDS, myelodysplastic syndrome; CML, chronic myeloid leukemia; AA, aplastic anemia; BM, bone marrow; PBSC, peripheral blood stem cell; CB, cord blood; CSP, cyclosporine; TAC, tacrolimus; sMTX, short-term methotrexate; MMF, mycophenolate mofetil; PTCy, posttransplant cyclophosphamide; ATG, anti-thymocyte globulin.

TABLE 2 Characteristics of acute GVHD

Total no. of patients	n = 200
Acute GVHD	128 (64.0%)
Grade II-IV	64 (32.0%)
Grade III-IV	25 (12.5%)
Organs involved in acute GVHD	
Skin only	90 (45.0%)
Intestine	36 (18.0%)
Liver	8 (4.0%)
Skin (stage 1/2/3/4)	41/35/32/4
Intestine (stage 1/2/3/4)	17/7/2/10
Liver (stage 1/2/3/4)	1/0/4/3
IV PSL or mPSL for acute GVHD	43 (21.5%)

GVHD, graft-vs -host disease; IV, intravenous injection; PSL, prednisolone; mPSL, methylprednisolone.

eradicate commensal bacteria that are critical for maintaining immune homeostasis in the intestine. Recent reports demonstrated that the use of prophylactic antibiotics was associated with severe acute GVHD and poor overall survival⁹ and that the use of broad-spectrum antibiotics as treatment for neutropenic fever early after allo-SCT led to an increased incidence of intestinal GVHD and GVHD-related mortality.¹⁰ We herein evaluated impacts of different antibiotic use on the incidence of intestinal GVHD after allo-SCT.

2 | PATIENTS AND METHODS

2.1 | Study design

Among 213 patients who underwent allo-SCT in our institution between January 2010 and December 2015, 200 cases with successful

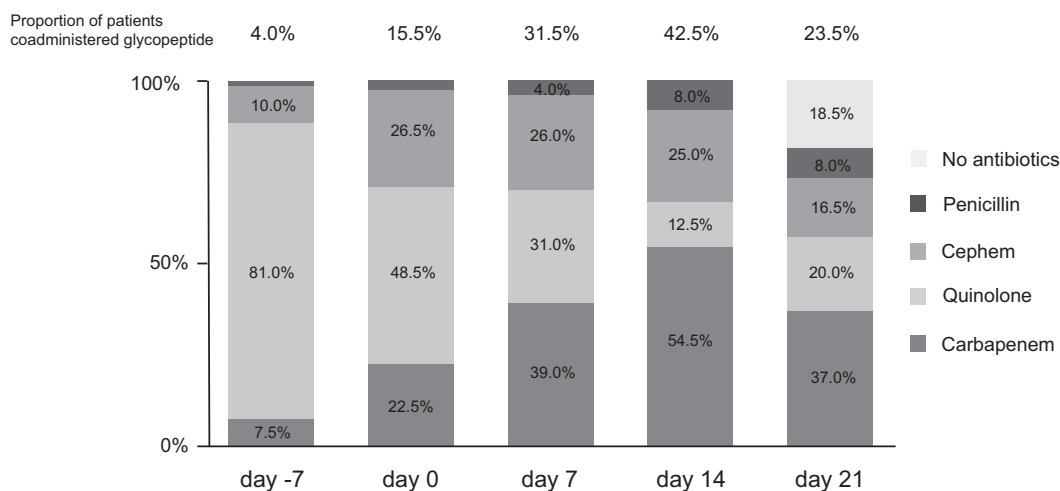


FIGURE 1 Proportion of used each antibiotics and patients coadministered glycopeptide on day -7, day 0, day 7, day 14, and day 21 after allo-SCT; Proportion of used each antibiotics: carbapenem, quinolone, cephem, penicillin, and no antibiotics. The upper percentage is the proportion of patients coadministered glycopeptide

engraftment were analyzed. Patients with graft failure ($n = 8$) and death before engraftment ($n = 5$) were excluded. Acute GVHD was diagnosed by clinical and/or pathological findings, and graded according to standard criteria.¹¹ Antibiotics were classified into carbapenem, quinolone, penicillin, cephem, and glycopeptide. Carbapenem included meropenem, doripenem, and biapenem. Quinolone included levofloxacin and pazufloxacin. Penicillin included piperacillin-tazobactam and piperacillin. Cephem included cefepime, ceftriaxone, and ceftazidime. Glycopeptide included teicoplanin and vancomycin. Oral antibiotics given for febrile neutropenia prophylaxis were also included in this analysis. The definition of bloodstream infection (BSI) was adapted from the 2004 Center for Disease Control and Prevention's definitions for nosocomial infections.¹² This study was approved by the Institutional Ethics Committee of Hokkaido University Hospital (approval number 017-0006) and conducted in compliance with the ethical principles based on the Declaration of Helsinki.

2.2 | Treatment and prophylaxis for bacterial infections

Febrile neutropenia prophylaxis was performed by oral quinolone. Febrile neutropenia was initially treated with fourth-generation cephalosporin in most of the cases. If not effective, antibiotics were switched to carbapenem or penicillin, or glycopeptide was added.

2.3 | Statistical analysis

Differences between two groups of data of continuous variables were analyzed by Mann-Whitney U test. Differences between two groups of data of proportions were analyzed by Fisher's exact test. We used Gray's test for the cumulative incidence of intestinal GVHD and nonrelapse mortality (NRM), and Fine and Gray's test for the univariate and multivariate analysis. The competing events that were included in Gray's test of cumulative incidence of intestinal GVHD were relapse and death without intestinal GVHD. The comparison of OS was analyzed by the logrank test. A Cox proportional hazard model was used to investigate survival rate. NRM was defined as any cause of death except for relapse or recurrence. OS was defined as a time from allo-SCT to death or last follow-up. P -values <0.05 were considered as a statistical significance. All analyses were performed with EZR (Saitama Medical Center, Jichi Medical University, Saitama, Japan).¹³

3 | RESULTS

3.1 | Patient characteristics

Patient characteristics are shown in Table 1. Ninety-one (45.5%), 64 (29.5%), and 45 (22.5%) patients underwent bone marrow transplantation (BMT), peripheral blood stem cell transplantation (PBSCT), and cord blood transplantation (CBT), respectively. Conditioning regimens were myeloablative in 90 (45.0%) patients

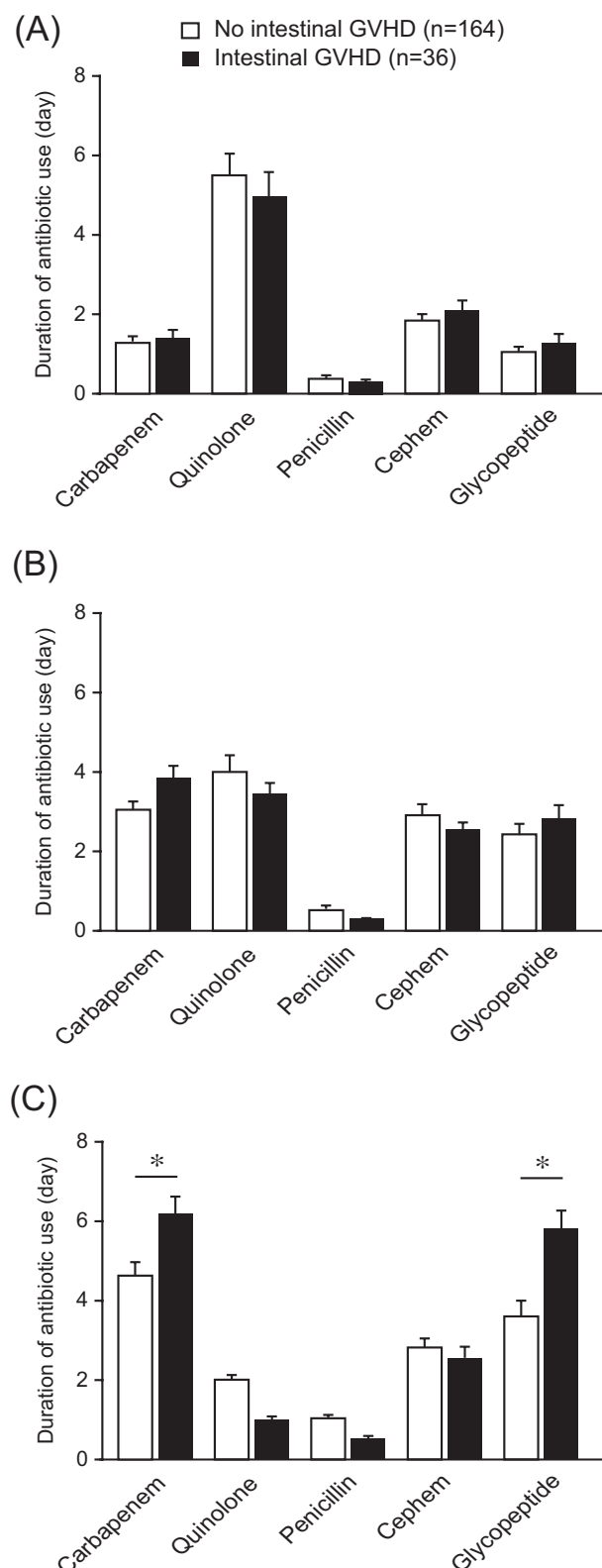


FIGURE 2 Comparison of antibiotic use in patients with intestinal GVHD and those without it in different 10-d periods. Data are shown as mean $\pm$ SD ($n = 200$). Open column: patients who did not develop intestinal GVHD ($n = 164$), filled column: patients who developed intestinal GVHD ($n = 36$); (A) day -7 to day +2, (B) day 0 to day +9, and (C) for 10 d up until 3 d after engraftment

and nonmyeloablative in 110 (55.0%) patients. One hundred thirty-six (68.0%) patients received calcineurin inhibitors and short-term methotrexate for GVHD prophylaxis. Twenty-one (10.5%) patients underwent HLA-haploidentical PBSCT using posttransplant cyclophosphamide. Thirty-nine (19.5%) patients had documented BSI before neutrophil engraftment, including 31 events caused by gram-positive bacteria and 8 events by gram-negative bacteria.

3.2 | Graft-vs-host disease characteristics

Characteristics of GVHD are shown in Table 2. One hundred twenty-eight patients (64.0%) developed acute GVHD: grade II to IV in 64 patients, and grade III to IV in 25 patients. Thirty-six patients (18.0%) presented with intestinal GVHD: isolated intestinal GVHD in 14 patients. Ten patients developed stage 4 intestinal GVHD. Forty-three patients (21.5%) were systemically treated with prednisolone or methylprednisolone against acute GVHD.

3.3 | Association between antibiotic use and the incidence of intestinal GVHD

Types of antibiotics used and proportion of patients coadministered glycopeptide on day -7, day 0, day 7, day 14, and day 21 after allo-SCT are shown in Figure 1. Quinolone was most used as febrile neutropenia prophylaxis on day -7. On the other hand, 19% of patients had already received antibiotics as treatment for bacterial infections on day -7. Use of carbapenem or penicillin increased gradually toward day 14. Glycopeptide was most used in combination on day 14. Antibiotic use was compared between patients with intestinal GVHD ($n = 36$) and those without it ($n = 164$, Figure 2). We evaluated association of intestinal GVHD and administration of antibiotics in different 10-day periods of antibiotic use: day -7 to day +2, day 0 to day +9, and for 10 days up until 3 days after engraftment. There was no difference in antibiotic use between patients with intestinal GVHD and those without it in the periods of day -7 to day +2 and day 0 to day +9 (Figure 2A, B). On the other hand, patients with intestinal GVHD received significantly longer administration of carbapenem

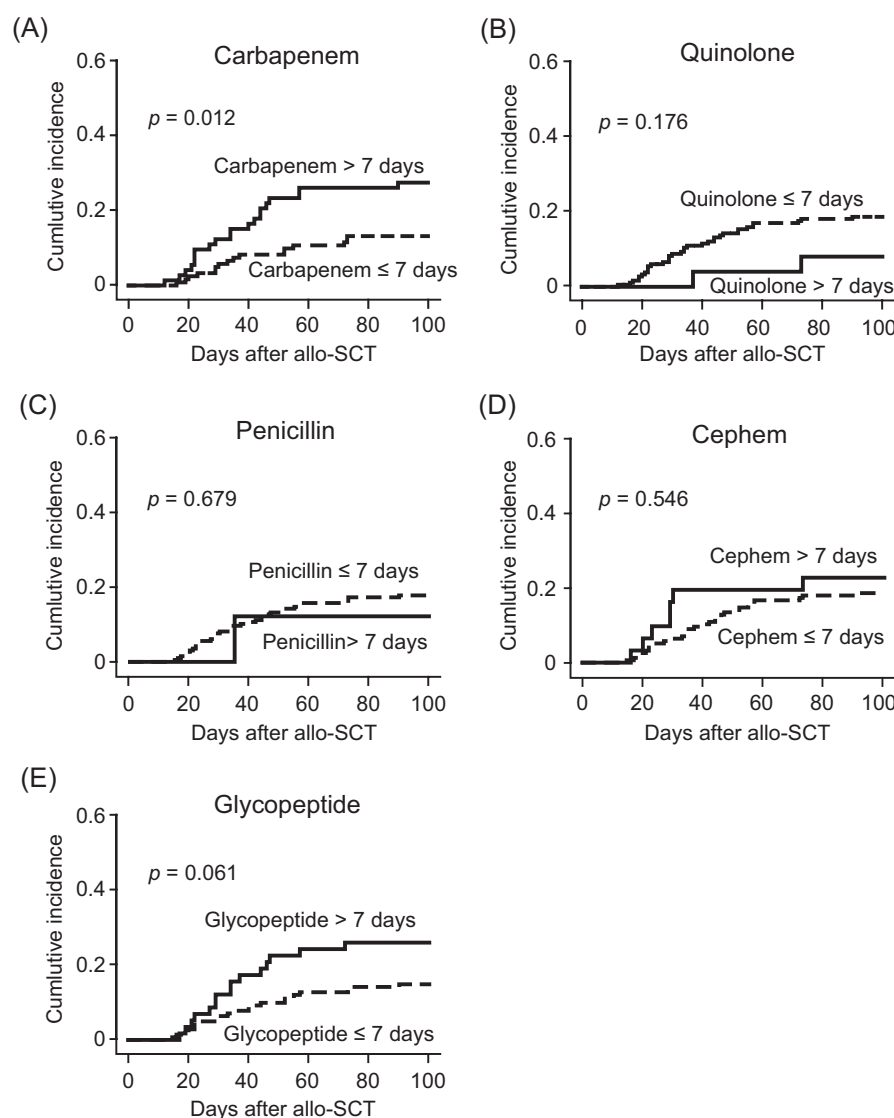


FIGURE 3 Cumulative incidence of intestinal GVHD according to each antibiotic use for greater than 7 d; (A) carbapenem use greater than for 7 d ($n = 75$) or not ($n = 125$), (B) quinolone use greater than for 7 d ($n = 24$) or not ($n = 176$), (C) penicillin use greater than for 7 d ($n = 8$) or not ($n = 192$), (D) cephem use greater than for 7 d ($n = 33$) or not ($n = 167$), and (E) glycopeptide use greater than for 7 d ($n = 58$) or not ($n = 142$)

TABLE 3 Characteristics of patients who received carbapenem for greater than 7 d and those who received carbapenem for 7 d or less

	Carbapenem > 7 d n = 75	Carbapenem ≤ 7 d n = 125	P-value
Age, median (range)	43 (17-67)	49 (19-67)	0.235
Male sex	50 (66.6%)	66 (56.9%)	0.057
Diagnosis			
AML	32 (42.7%)	43 (34.4%)	0.291
ALL/LBL	18 (24.0%)	23 (18.4%)	0.369
ML/ATLL	10 (13.3%)	32 (25.6%)	0.052
MDS	11 (14.7%)	15 (12.0%)	0.665
CML	2 (2.7%)	3 (2.4%)	1.000
AA	1 (1.3%)	3 (2.4%)	1.000
Graft source			
BM match	14 (18.7%)	43 (34.4%)	0.023
BM mismatch	10 (13.3%)	24 (19.2%)	0.334
PBSC match	12 (16.0%)	13 (10.4%)	0.274
PBSC mismatch	21 (28.0%)	18 (14.4%)	0.026
CB	18 (24.0%)	23 (18.4%)	0.369
Conditioning regimen			
Myeloablative	38 (50.7%)	52 (41.6%)	0.249
Reduced intensity	37 (49.3%)	73 (58.4%)	
GVHD prophylaxis			
CSP/TAC + sMTX	43 (57.3%)	93 (74.4%)	0.018
CSP/TAC + MMF	7 (9.3%)	9 (7.2%)	0.600
CSP/TAC + MMF + PTCy	12 (16.0%)	9 (7.2%)	0.059
CSP/TAC alone	10 (13.3%)	8 (6.4%)	0.126
CSP/TAC + sMTX + ATG	3 (4.0%)	6 (4.8%)	1.000

AML, acute myeloid leukemia; ALL, acute lymphoblastic leukemia; LBL, lymphoblastic leukemia; ML, malignant lymphoma; ATLL, adult T-cell leukemia/lymphoma; MDS, myelodysplastic syndrome; CML, chronic myeloid leukemia; AA, aplastic anemia; BM, bone marrow; PBSC, peripheral blood stem cell; CB, cord blood; CSP, cyclosporine; TAC, tacrolimus; sMTX, short-term methotrexate; MMF, mycophenolate mofetil; PTCy, posttransplant cyclophosphamide; ATG, anti-thymocyte globulin.

TABLE 4 Univariate and multivariate analysis for the incidence of intestinal GVHD

Variable	Univariate analysis		Multivariate analysis	
	HR (95% CI)	P-value	HR (95% CI)	P-value
Carbapenem > 7 d	2.28 (1.19-4.37)	0.013	2.75 (1.39-5.45)	0.004
BSI	1.23 (0.56-2.70)	0.610	1.27 (0.52-3.10)	0.600
BMT	0.95 (0.49-1.82)	0.870	1.00	-
PBSCT	0.69 (0.33-1.47)	0.340	0.76 (0.30-1.92)	0.570
CBT	1.59 (0.79-3.22)	0.190	1.31 (0.56-3.06)	0.540
HLA mismatch	1.39 (0.70-2.78)	0.350	1.51 (0.59-3.87)	0.390
GVHD prophylaxis except CSP/TAC + sMTX	0.72 (0.33-1.58)	0.410	0.51 (0.20-1.28)	0.150

GVHD, graft-vs-host disease; HR, hazard ratio; CI, confidence interval; BSI, bloodstream infection; BMT, bone marrow transplantation; PBSCT, peripheral blood stem cell transplantation; CBT, cord blood transplantation; HLA, human leukocyte antigen; CSP, cyclosporine; TAC, tacrolimus; sMTX, short-term methotrexate.

(mean 4.6 vs 6.2 days; $P = 0.045$) and glycopeptide (mean 3.7 vs 5.9 days; $P = 0.004$) compared to those without it in the periengraftment period (Figure 2C). All 200 patients received antibiotics in the periengraftment period: 181 (90.5%) patients for treatment and 19 (9.5%) patients for prophylaxis.

We determined the cut-off value of duration of antibiotic use by the ROC curve analysis. Use for more than 7 days was most predictive with specificity of 0.665 and sensitivity of 0.556 for carbapenem (Figure S1A), and specificity of 0.841 and sensitivity of 0.194 for cephem (Figure S1B). As glycopeptide was always used in combination with other antibiotics and number of patients receiving quinolone or penicillin was too small, cut-off was set to 7 days according to the results of carbapenem and cephem. Patients were divided into those who received antibiotics for greater than 7 days

and those who received antibiotics for 7 days or less. The cumulative incidence of intestinal GVHD stratified by use of antibiotics for greater than 7 days is shown in Figure 3. Administration of carbapenem for greater than 7 days was associated with higher incidence of intestinal GVHD ($P = 0.012$, Figure 3A), while use of other antibiotics was not significantly associated with the incidence of intestinal GVHD (Figure 3B-E). A previous report showed that use of imipenem-cilastatin or piperacillin-tazobactam was associated with higher incidence of intestinal GVHD, suggesting that anaerobic bacterium plays an important role to regulate immune reaction.¹⁰ Our data showed that the administration of penicillin for greater than 7 days did not significantly increase the incidence of intestinal GVHD, although a number of patients were too small to detect any differences (Figure 3C). Characteristics of patients who received

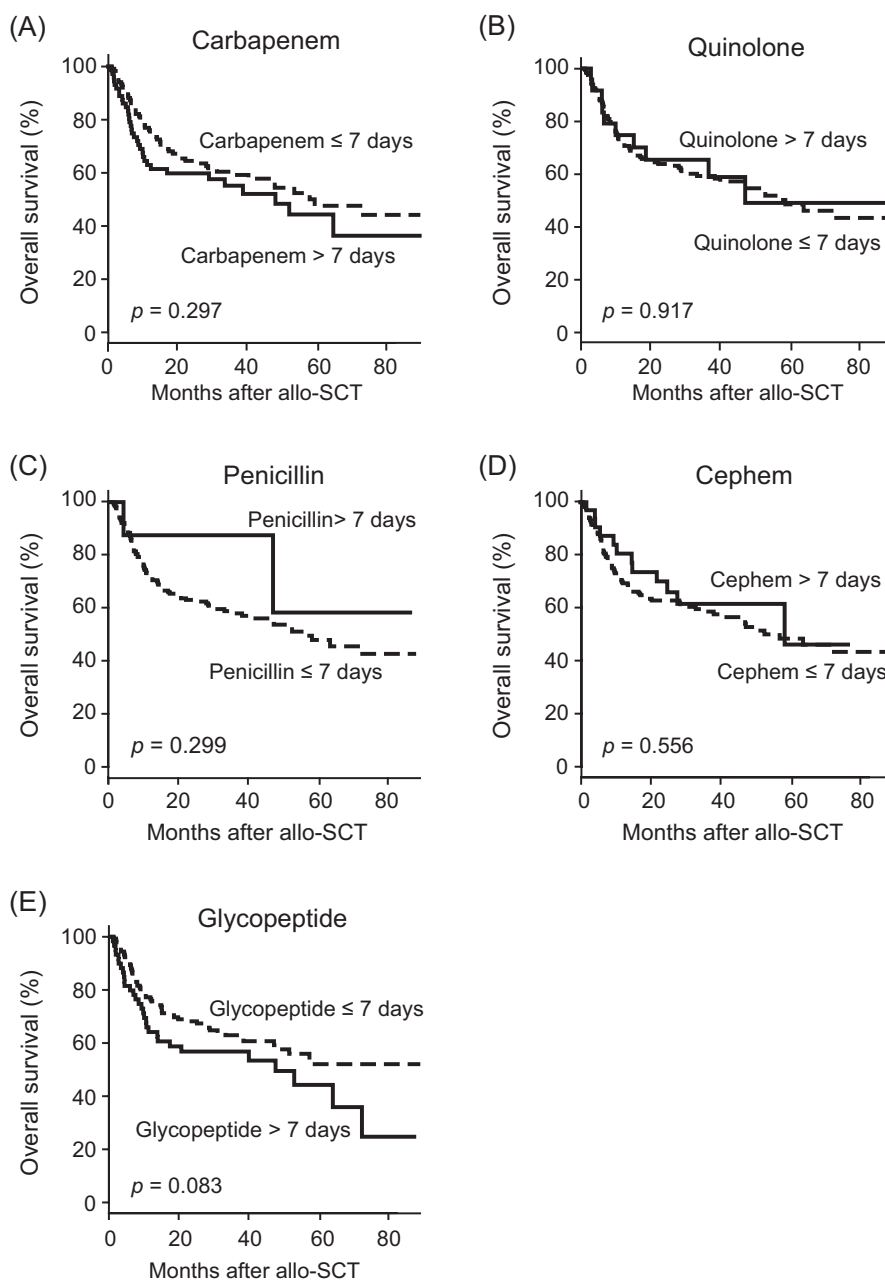


FIGURE 4 OS in patients who received each antibiotic use for greater than 7 d; (A) carbapenem use greater than for 7 d ($n = 75$) or not ($n = 125$), (B) quinolone use greater than for 7 d ($n = 24$) or not ($n = 176$), (C) penicillin use greater than for 7 d ($n = 8$) or not ($n = 192$), (D) cephem use greater than for 7 d ($n = 33$) or not ($n = 167$), and (E) glycopeptide use greater than for 7 d ($n = 58$) or not ($n = 142$)

carbapenem for greater than 7 days and those who received carbapenem for 7 days or less is shown in Table 3. In patients given carbapenem for greater than 7 days, there were significantly many patients donated from PBSC mismatch donors, while there were significantly few patients donated from BM match donor and receiving calcineurin + short-term MTX as GVHD prophylaxis compared to those given carbapenem for 7 days or less. In multivariate analysis, use of carbapenem for greater than 7 days (hazard ratio 2.75, 95% confidence interval 1.39-5.45, $P = 0.004$) was significantly associated with an increased incidence of intestinal GVHD. BSI, donor sources, HLA mismatch, and GVHD prophylaxis were not associated with the incidence of intestinal GVHD (Table 4).

3.4 | OS and NRM

OS and NRM with a median follow-up of 807 days, ranging from 24 to 2712 days in patients with each antibiotic use, were showed in

Figure 4A-E and Figure 5A-E. Use of antibiotics in periengraftment period for greater than 7 days was not associated with poor OS and high NRM.

4 | DISCUSSION

Intestinal GVHD is associated with abnormal intestinal microbiota, dysbiosis. Administration of broad-spectrum antibiotics disturbs composition and diversity of intestinal microbiota.^{14,15} Recent studies indicated that intestinal microbiota at engraftment was associated with acute GVHD.^{16,17} In our study cohort, patients with intestinal GVHD more received broad-spectrum antibiotics in periengraftment period. It was possible that intestinal microbiota of patients with intestinal GVHD was more disturbed. Emergence of dysbiosis is associated with poor survival after allo-SCT.^{2,5,18} A recent study demonstrated that the use of imipenem-cilastatin or

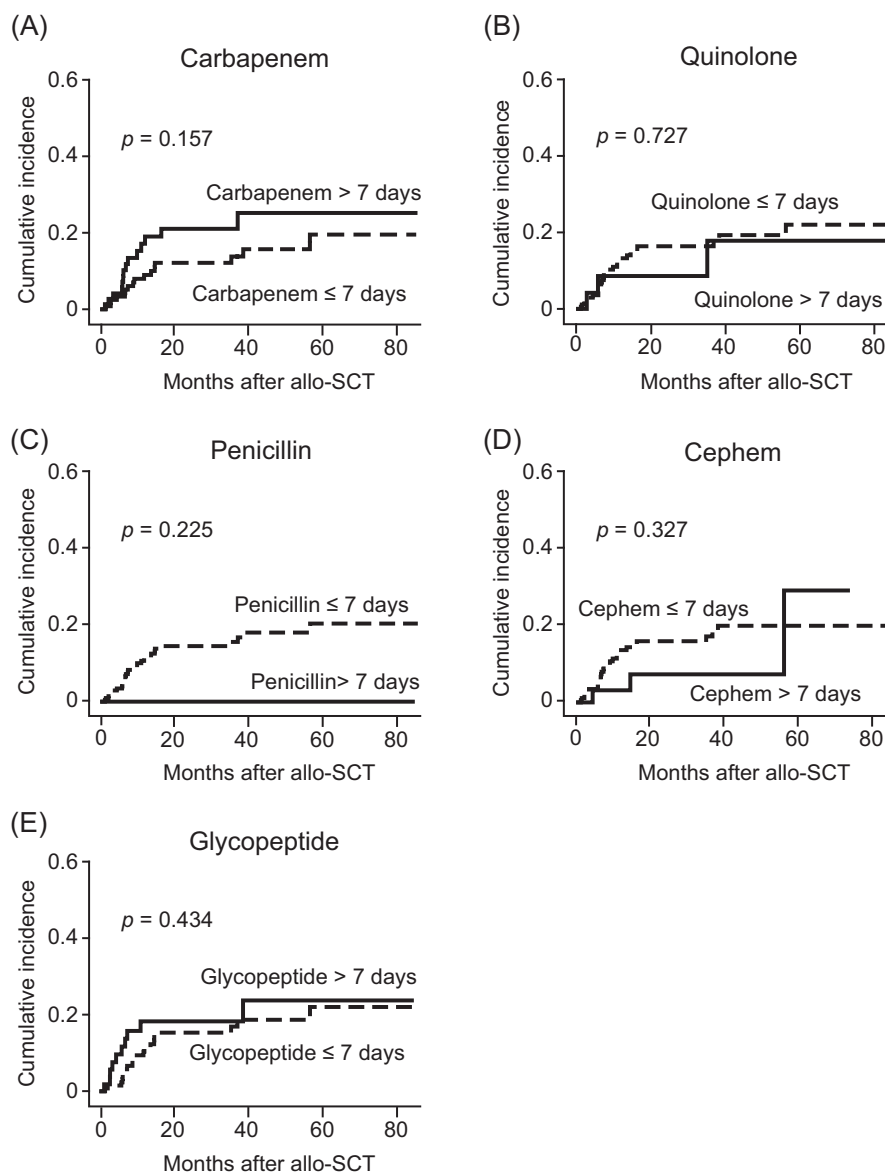


FIGURE 5 NRM in patients who received each antibiotic use for greater than 7 d; (A) carbapenem use greater than for 7 d ($n = 75$) or not ($n = 125$), (B) quinolone use greater than for 7 d ($n = 24$) or not ($n = 176$), (C) penicillin use greater than for 7 d ($n = 8$) or not ($n = 192$), (D) cephem use greater than for 7 d ($n = 33$) or not ($n = 167$) and (E) glycopeptide use greater than for 7 d ($n = 58$) or not ($n = 142$)

piperacillin-tazobactam for neutropenic fever early after allo-SCT led to an increase incidence of intestinal GVHD and GVHD-related mortality.¹⁰ Our study also showed that use of carbapenem for greater than 7 days was a risk for intestinal GVHD. On the other hand, use of carbapenem for greater than 7 days was not associated with poor OS and high NRM. Farowski et al¹⁹ reported that cumulative antibiotic exposure was related to the incidence of acute GVHD. Long-term administration of carbapenem may cause loss of beneficial anaerobic bacteria for regulating alloreactive response, such as *Clostridia* and *Blautia*. In mice, *Clostridia* induce regulatory T cells in the intestine and maintain immune homeostasis.²⁰ A recent study demonstrated that intestinal *Blautia* abundance can be used as a good prognostic indicator for GVHD-related mortality.²¹

A recent study demonstrated that prophylactic use of antibiotics for gut decontamination was associated with severe acute GVHD involving gastrointestinal organs and poor OS.⁹ Other study showed that gut decontamination with rifaximin was associated with lower transplant-related mortality, higher OS, lower incidence of intestinal GVHD, compared to decontamination using ciprofloxacin or metronidazole.²² Most of antibiotics for prophylaxis in this study were oral quinolone, and the duration of prophylactic use was short. There was no significant impact of the use of oral quinolone on the incidence of intestinal GVHD. Previous studies showed an association between BSI and an increased occurrence of grade II-IV acute GVHD.^{23,24} However, in our study, such an association was not found. Peled *et al* reported that the abundance of certain intestinal bacteria was associated with reduction of relapse after allo-SCT without an increased incidence of GVHD.⁷ Modulation of intestinal microbiota by prebiotics, probiotics, antibiotics, fecal microbiota transplant, and Paneth cell growth factor R-Spondin 1 may represent novel approach to improve transplant outcome.^{4,14,25,26}

This study has several limitations, as this is a retrospective analysis and antibiotic use is decided by physician's preference. Therefore, our results should not guide choice of antibiotics. In conclusion, long use of carbapenem for periengraftment period was associated with the incidence of intestinal GVHD. Prospective studies are required to confirm observations and establish new approach by intervention to intestinal microbiota.

CONFLICT OF INTEREST

The authors declare no competing financial interests.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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